

Question 1

PART A (45 marks)

Activity:

Load the *employee_performance_dataset.csv* file into WEKA.

The dataset is used to classify employee performance into "High" or "Low" categories to assist the Human Resource department in identifying top talent for promotion and identifying employees who need additional training.

QUESTION 1

a) Record the total number of **instances** and **attributes** in the dataset.

Provide a screenshot of the WEKA Preprocess panel showing this information.

Number of Instances	100
Number of Attributes	8

Current relation	
Relation: employee_performance_dataset	Attributes: 8
Instances: 100	Sum of weights: 100

Question 2

Essay (Total Point: 2.00)

Question 2

Duration

b) Identify the target **class attribute** and list at least **THREE (3) numeric** predictors found in the dataset.

Class attribute : Performance

Numeric:

- 1) Attendance
- 2) Training_Hours
- 3) Task_Completion

Relation: employee_performance_dataset								
No.	1: Employee_ID	2: Attendance	3: Training_Hours	4: Task_Completion	5: Experience	6: Overtime	7: Distance_From_Office	8: Performance
	Nominal	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	Nominal

Question 3

Essay (Total Point: 2.00)

Question 3

Duration

c) Scan the dataset for **missing values**. If missing values were present, **state** which WEKA **filter** would be used to handle them.

No missing values found. If found, choose filters – unsupervised – attribute – ReplaceMissingValues.

Employee_ID	Selected attribute Name: Employee_ID Missing: 0 (0%)
Attendance	Selected attribute Name: Attendance Missing: 0 (0%)
Training_Hours	Selected attribute Name: Training_Hours Missing: 0 (0%)
Task_Completion	Selected attribute Name: Task_Completion Missing: 0 (0%)
Experience	Selected attribute Name: Experience Missing: 0 (0%)
Overtime	Selected attribute Name: Overtime Missing: 0 (0%)
Distance_From_Office	Selected attribute Name: Distance_From_Office Missing: 0 (0%)
Perfomance	Selected attribute Name: Performance Missing: 0 (0%)

Question 4

Question 4

Duration

d) Examine the **summary statistics** of the **numeric attributes** in WEKA. Record the **minimum**, **maximum**, and **mean** values for any **THREE (3)** numeric attributes. **Explain** briefly how these statistics help in understanding the dataset before building classification models.

Insert screenshot of the WEKA Preprocess panel showing the statistics for the selected attributes.

Attribute	Minimum	Maximum	Mean
Attendance	62	98	79.92
Training_Hours	2	22	11.59
Task_Completion	58	99	80.18

Explanation:-

These numeric attributes helps in separate the instances with value given.

Attendance

Selected attribute	
Name: Attendance	Type: Numeric
Missing: 0 (0%)	Distinct: 34
	Unique: 9 (9%)
Statistic	Value
Minimum	62
Maximum	98
Mean	79.92
StdDev	10.089

Training_Hours

Selected attribute	
Name: Training_Hours	Type: Numeric
Missing: 0 (0%)	Distinct: 21
	Unique: 1 (1%)
Statistic	Value
Minimum	2
Maximum	22
Mean	11.59
StdDev	5.305

Task_Completion

Selected attribute	
Name: Task_Completion	Type: Numeric
Missing: 0 (0%)	Unique: 6 (6%)
Distinct: 37	
Statistic	Value
Minimum	58
Maximum	99
Mean	80.18
StdDev	11.587

Question 5

Essay (Total Point: 4.00)

Question 5

Duration

QUESTION 2

a) Apply **10-fold Cross-Validation** to build **J48** and **Random Forest** classifiers in WEKA. Include screenshots of the classification outputs.

J48:

```
Classifier output
=== Run information ===

Scheme:      weka.classifiers.trees.J48 -C 0.25 -M 2
Relation:    employee_performance_dataset
Instances:   100
Attributes:  8
             Employee_ID
             Attendance
             Training_Hours
             Task_Completion
             Experience
             Overtime
             Distance_From_Office
             Performance
Test mode:   10-fold cross-validation

=== Classifier model (full training set) ===

J48 pruned tree
-----
Attendance <= 80
| Attendance <= 78: Low (42.0)
| Attendance > 78
| | Training_Hours <= 7: Low (4.0)
| | Training_Hours > 7
| | | Experience <= 6: High (3.0)
| | | Experience > 6: Low (4.0/1.0)
Attendance > 80: High (47.0/1.0)

Number of Leaves :    5
Size of the tree :    9
```

```

Time taken to build model: 0.01 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances      90          90    %
Incorrectly Classified Instances    10          10    %
Kappa statistic                    0.8
Mean absolute error                0.1153
Root mean squared error            0.3153
Relative absolute error            23.0534 %
Root relative squared error        63.0597 %
Total Number of Instances         100

=== Detailed Accuracy By Class ===

                TP Rate  FP Rate  Precision  Recall   F-Measure  MDC     ROC Area  PRC Area  Class
                0.900   0.100   0.900     0.900   0.900     0.800   0.876    0.808    High
                0.900   0.100   0.900     0.900   0.900     0.800   0.876    0.887    Low
Weighted Avg.   0.900   0.100   0.900     0.900   0.900     0.800   0.876    0.848

=== Confusion Matrix ===
  a  b  <-- classified as
45  5 | a = High
 5 45 | b = Low

```

RandomForest:

```

Classifier output

=== Run information ===

Scheme:      weka.classifiers.trees.RandomForest -P 100 -I 100 -num-slots 1 -K 0 -M 1.0 -V 0.001 -S 1
Relation:    employee_performance_dataset
Instances:    100
Attributes:   8
              Employee_ID
              Attendance
              Training_Hours
              Task_Completion
              Experience
              Overtime
              Distance_From_Office
              Performance
Test mode:    10-fold cross-validation

=== Classifier model (full training set) ===

RandomForest

Bagging with 100 iterations and base learner

weka.classifiers.trees.RandomTree -K 0 -M 1.0 -V 0.001 -S 1 -do-not-check-capabilities

Time taken to build model: 0.05 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances      93          93    %
Incorrectly Classified Instances     7           7    %
Kappa statistic                    0.86
Mean absolute error                0.2788
Root mean squared error            0.2969
Relative absolute error            55.7576 %
Root relative squared error        59.3787 %
Total Number of Instances         100

```

```

=== Detailed Accuracy By Class ===

                TP Rate  FP Rate  Precision  Recall   F-Measure  MDC     ROC Area  PRC Area  Class
                0.900   0.040   0.957     0.900   0.928     0.862   0.991    0.992    High
                0.960   0.100   0.906     0.960   0.932     0.862   0.991    0.991    Low
Weighted Avg.   0.930   0.070   0.932     0.930   0.930     0.862   0.991    0.992

=== Confusion Matrix ===
  a  b  <-- classified as
45  5 | a = High
 2 48 | b = Low

```

Question 6

Essay (Total Point: 3.00)

[Question 6](#)

Duration

b) **Compare** the predictive performance of **J48** and **Random Forest** by reporting the **percentage of correctly classified instances** and **determine** which algorithm performs better.

Classifier	Accuracy(%)
------------	-------------

J48	90
Random Forest	93

RandomForest performs better as it produce higher accuracy which are 93% compare to J48 90%.

Question 7

Essay (Total Point: 4.00)

Question 7

Duration

c) Apply the **Percentage Split (66%)** test option in WEKA and report the **Accuracy**, **True Positive (TP)**, **True Negative (TN)**, **False Positive (FP)**, and **False Negative (FN)** values for **J48** and **Random Forest** classifiers.

Classifier	Accuracy(%)	TP	TN	FP	FN
J48	94.1176	17	15	2	0
Random Forest	91.1765	15	16	1	2

Question 8

Essay (Total Point: 1.00)

Question 8

Duration

d) Using the **J48 model**, identify the **Predicted Class** for the following instances.

Instance No.	Actual Class	Predicted Class
Employee E003	2:Low	1:High
Employee E012	1:High	1:High
Employee E034	2:Low	2:Low

=== Predictions on test split ===				
inst#	actual	predicted	error	prediction
1	1:High	1:High		1
2	1:High	1:High		1
3	2:Low	1:High	+	1
4	2:Low	2:Low		1
5	2:Low	2:Low		1
6	2:Low	2:Low		1
7	1:High	1:High		1
8	2:Low	2:Low		1
9	1:High	1:High		1
10	1:High	1:High		1
11	2:Low	2:Low		1
12	1:High	1:High		1
13	2:Low	2:Low		1
14	1:High	1:High		1
15	2:Low	1:High	+	1
16	1:High	1:High		1
17	1:High	1:High		1
18	1:High	1:High		1
19	2:Low	2:Low		1
20	1:High	1:High		1
21	2:Low	2:Low		1
22	2:Low	2:Low		1
23	2:Low	2:Low		1
24	2:Low	2:Low		1
25	2:Low	2:Low		1
26	1:High	1:High		1
27	1:High	1:High		1
28	1:High	1:High		1
29	1:High	1:High		1
30	1:High	1:High		1
31	2:Low	2:Low		1
32	2:Low	2:Low		1
33	1:High	1:High		1
34	2:Low	2:Low		1

Question 9

Essay (Total Point: 1.00)

[Question 9](#)

Duration

QUESTION 3

a) **Visualize** the generated **J48 tree** and answer the following:

i. Which attribute serves as the **root node**?

Root Node: Attendance

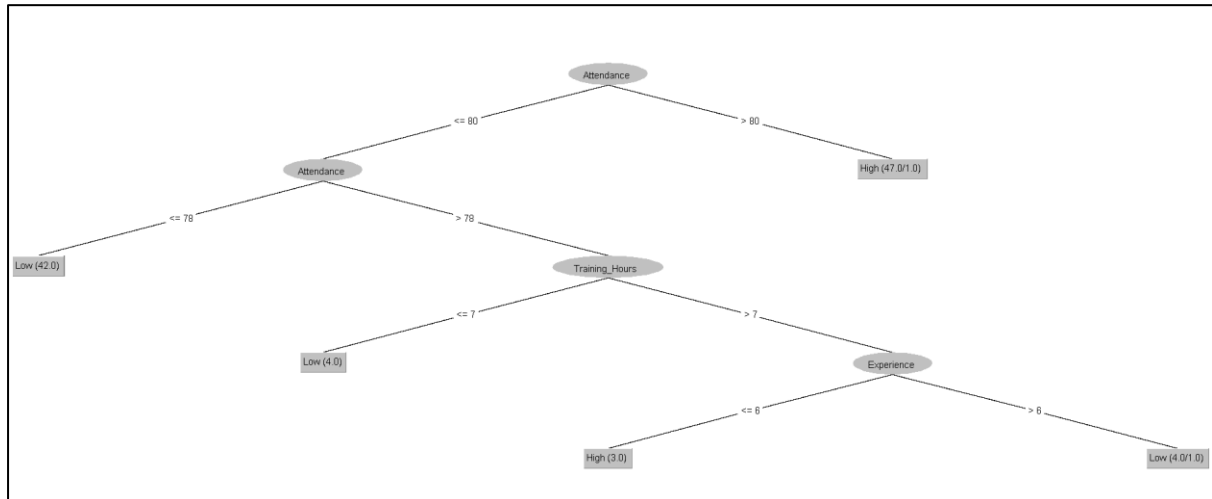
Question 10

Essay (Total Point: 4.00)

Question 10

Duration

ii. Draw the **tree** generated by WEKA for this dataset.



Question 11

Essay (Total Point: 3.00)

Question 11

Duration

b) Derive and write any **THREE (3) IF-THEN** classification rules from the generated **J48 decision tree**.

Rules:

- 1) IF Attendance > 80
THEN Performance = High
- 2) IF Attendance <= 80 AND Attendance <= 78
THEN Performance = Low
- 3) IF Attendance <= 80 AND Attendance > 78 AND Training_Hours <= 7
THEN Performance = Low

Question 12

Essay (Total Point: 3.00)

Question 12

Duration

c) Using the **InfoGainAttributeEval** ranking produced by WEKA, list the top **THREE (3)** attributes that most significantly influence employee performance.

- 1) Employee_ID
- 2) Attendance
- 3) Task_Completion


```

Ranked attributes:
1.00000000000000002      1 Employee_ID
0.8261591059916662      2 Attendance
0.7255917221956928      4 Task_Completion
0.42739771881424904      3 Training_Hours
0                          6 Overtime
0                          5 Experience
0                          7 Distance_From_Office

Selected attributes: 1,2,4,3,6,5,7 : 7

```

Question 13

Essay (Total Point: 7.00)

Question 13

Duration

QUESTION 4

a) Generate a **reduced dataset** by selecting the most important attributes identified using **InfoGainAttributeEval** and rebuild the **J48 classifier**. Compare the **Accuracy**, **Error Rate**, **TP**, **TN**, **FP**, and **FN** values. Show the **calculations** for **Accuracy**, **Error Rate**, **Precision**, and **Recall**.

Accuracy(%)	Error Rate	TP	TN	FP	FN
94.1176	5.8824	16	16	1	1

Precision (Class a)	0.941	Recall (Class a)	0.941
Precision (Class b)	0.941	Recall (Class b)	0.941

Formula:

$$\text{Accuracy} = [\text{TP} + \text{TN} / \text{TP} + \text{TN} + \text{FP} + \text{FN}] * 100$$

$$\text{Error Rate} = [\text{FN} + \text{FP} / \text{TP} + \text{TN} + \text{FP} + \text{FN}] * 100$$

$$\text{Precision} = \text{TP} / \text{P} \quad \text{TN} / \text{N}$$

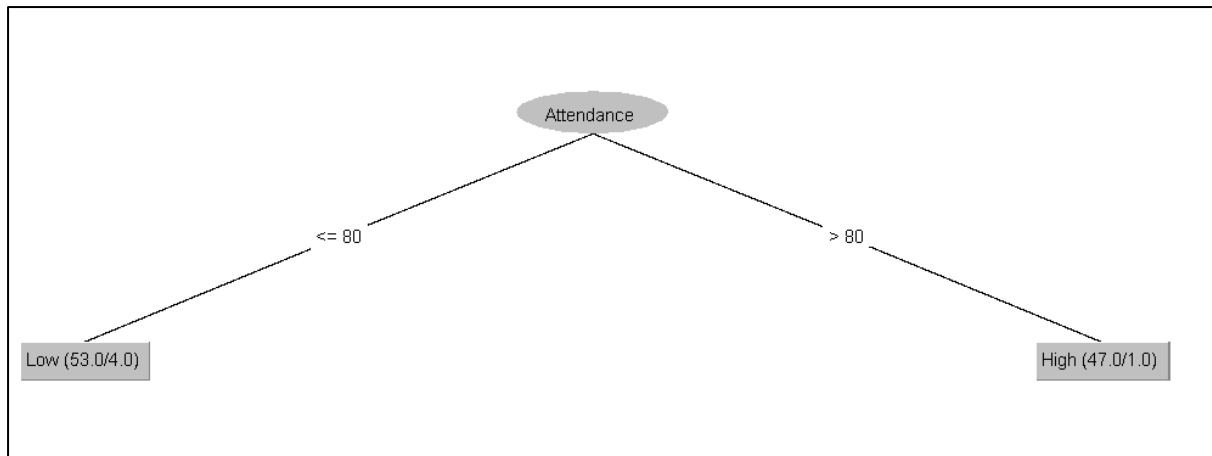
$$\text{Recall} = \text{TP} / \text{TP} + \text{FN} \quad \text{TN} / \text{FP} + \text{TN}$$

Question 14

Question 14

Duration

b) Draw the **tree** generated from **reduced dataset** and compare the **accuracy** and **tree complexity** with the original model.



Reduced dataset use less attribute that will make it more accuracy. Also it has fewer branch, node and leaf that will make the tree less complex and easier to understand

Question 15

Load the *customer_segmentation_dataset.csv* file into WEKA and perform the following tasks.

QUESTION 5

a) Remove the **Customer_ID** attribute and **state why** it should be **removed** before clustering.

Current relation
 Relation: customer_segmentation_dataset-weka.filters.unsupervised.attribute.Remove-R1
 Instances: 15
 Attributes: 3
 Sum of weights: 15

Attributes

All None Invert Pattern

No.	Name
1	☐ Age
2	☐ Annual_Income_k
3	☐ Spending_Score

Explanation:

Customer_ID is useless as the clustering does not need the class label. It use to clustering the instance to similar characteristic.

Question 16

Essay (Total Point: 1.00)

Question 16

Duration

b) Apply **SimpleKMeans** with **k = 3** and **report**:

i. Number of clusters generated

Clustered Instances

0	5 (33%)
1	5 (33%)
2	5 (33%)

Number of clusters generated : 3

Question 17

Essay (Total Point: 1.00)

Question 17

Duration

ii. Cluster sizes

Cluster sizes : 15

Question 18

Essay (Total Point: 1.00)

Question 18

Duration

iii. Based on the cluster centroids, briefly describe **one characteristic** of any **ONE** cluster.

Final cluster centroids:

Attribute	Full Data (15.0)	Cluster#		
		0 (5.0)	1 (5.0)	2 (5.0)
Age	40.0667	25.2	40	55
Annual_Income_k	49.2	22.6	50	75
Spending_Score	48.3333	80	50	15

Explanation:

Follow the range of the attribute which give similar characteristics and make it to the cluster group.